

Project Report: CPU Scheduling Algorithms Comparison

1. Project Overview

This project is a specialized simulation tool designed to study and compare the behavior of different CPU Scheduling algorithms, specifically **Shortest Job First (SJF)** and **Priority Scheduling**. The system provides a graphical interface to visualize how the CPU manages multiple processes, calculates performance metrics, and handles both **Non-Preemptive** and **Preemptive** (SRTF/Priority Preemptive) modes.

The core objective is to analyze the trade-offs between system efficiency (minimizing wait times) and task urgency (prioritizing critical jobs).

2. Key Performance Metrics

To evaluate the algorithms, the following metrics are calculated for each process:

- **Arrival Time (AT):** The time at which the process enters the ready queue.
- **Burst Time (BT):** The actual CPU time required by the process.
- **Waiting Time (WT):** The total time a process spends waiting in the ready queue ($WT = TAT - BT$).
- **Turnaround Time (TAT):** The total time from arrival to completion ($TAT = \text{Finish Time} - \text{Arrival Time}$).
- **Response Time (RT):** The time from arrival until the first time the process gets the CPU.

3. Test Scenarios & Analysis

Scenario A: Basic Mixed Workload

- **Goal:** To test the standard behavior of algorithms with varied arrival and burst times.

- **Analysis:** SJF prioritized shorter tasks, leading to the lowest average waiting time. Priority scheduling followed the assigned ranks, which caused a slight increase in the overall waiting time.
- **Conclusion:** SJF is optimal for reducing average delays in a normal workload.

SJF vs Priority Scheduling Project

SJF vs Priority Comparison

Input
Gantt Charts
Results
Analysis & Conclusion

Scheduling Mode

☐ SJF Preemptive (SRTF)
☐ Priority Preemptive

Process Input Table

PID	Arrival Time	Burst Time	Priority
P1	0	8	2
P2	1	4	1
P3	2	9	3
P4	3	5	4
P5	4	2	5

+ Add Process
Remove Selected
Clear All
Run Simulation



SJF vs Priority Scheduling Project

SJF vs Priority Comparison

Input | Gantt Charts | Results | Analysis & Conclusion

SJF — Average Metrics

Avg WT	Avg TAT	Avg RT
8.20	13.80	8.20

Priority — Average Metrics

Avg WT	Avg TAT	Avg RT
11.40	17.00	11.40

SJF — Per-Process Metrics

PID	AT	BT	Priority	FT	WT	TAT	RT
P1	0	8	2	8	0	8	0
P5	4	2	5	10	4	6	4
P2	1	4	1	14	9	13	9
P4	3	5	4	19	11	16	11
P3	2	9	3	28	17	26	17

Priority — Per-Process Metrics

PID	AT	BT	Priority	FT	WT	TAT	RT
P1	0	8	2	8	0	8	0
P2	1	4	1	12	7	11	7
P3	2	9	3	21	10	19	10
P4	3	5	4	26	18	23	18
P5	4	2	5	28	22	24	22

★ Comparison Summary

Average Waiting Time:	SJF: 8.20	Priority: 11.40	Winner: ✓ SJF
Average Turnaround Time:	SJF: 13.80	Priority: 17.00	Winner: ✓ SJF
Average Response Time:	SJF: 8.20	Priority: 11.40	Winner: ✓ SJF
SJF max single-process wait : 17 units			
Priority max single-process wait : 22 units			

SJF vs Priority Scheduling Project

SJF vs Priority Comparison

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Gantt Charts

Results

Analysis & Conclusion

Comparative Scheduling Analysis

Waiting & Response Time

- SJF average waiting time: **8.20** | Priority: **11.40** → Winner: **SJF**
- SJF achieved the **lowest average waiting time** by always selecting the shortest burst.
- SJF average response time: **8.20** | Priority: **11.40** → Winner: **SJF**

Turnaround Time

- SJF average TAT: **13.80** | Priority: **17.00** → Winner: **SJF**
- SJF provided **shorter turnaround time** — short jobs finish and leave quickly.

Fairness & Starvation Risk

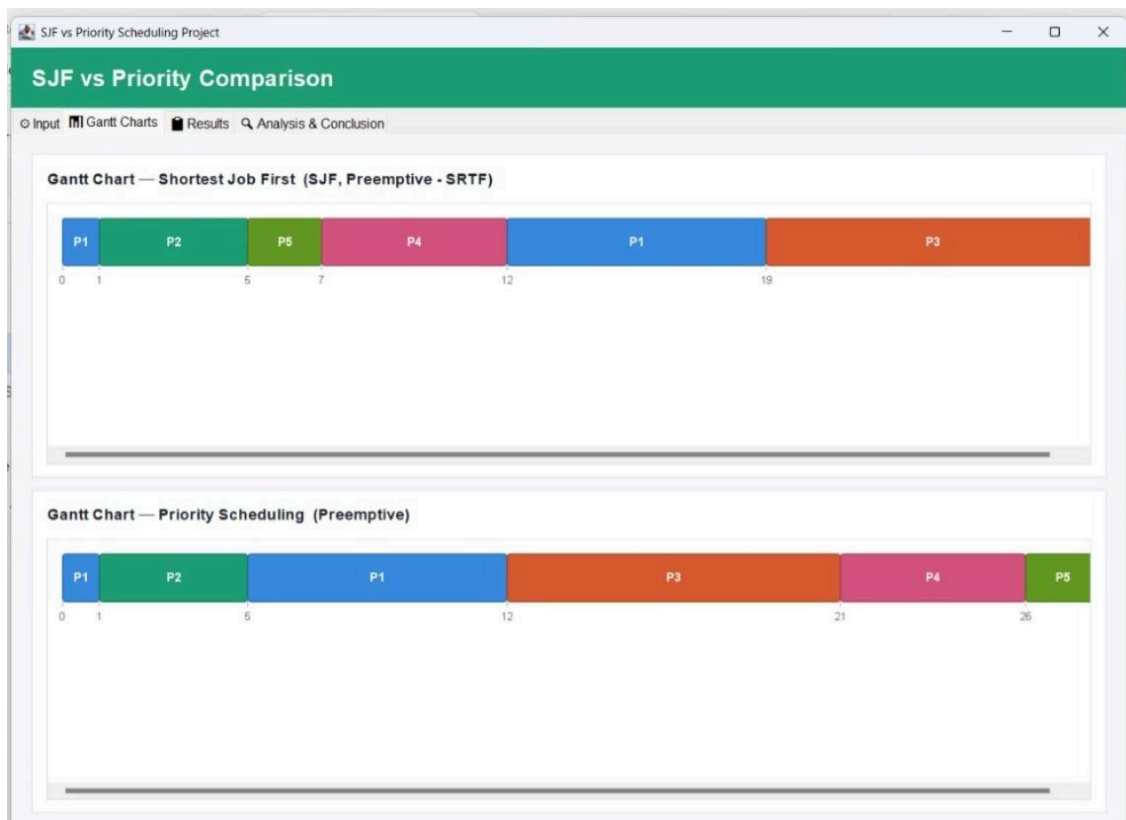
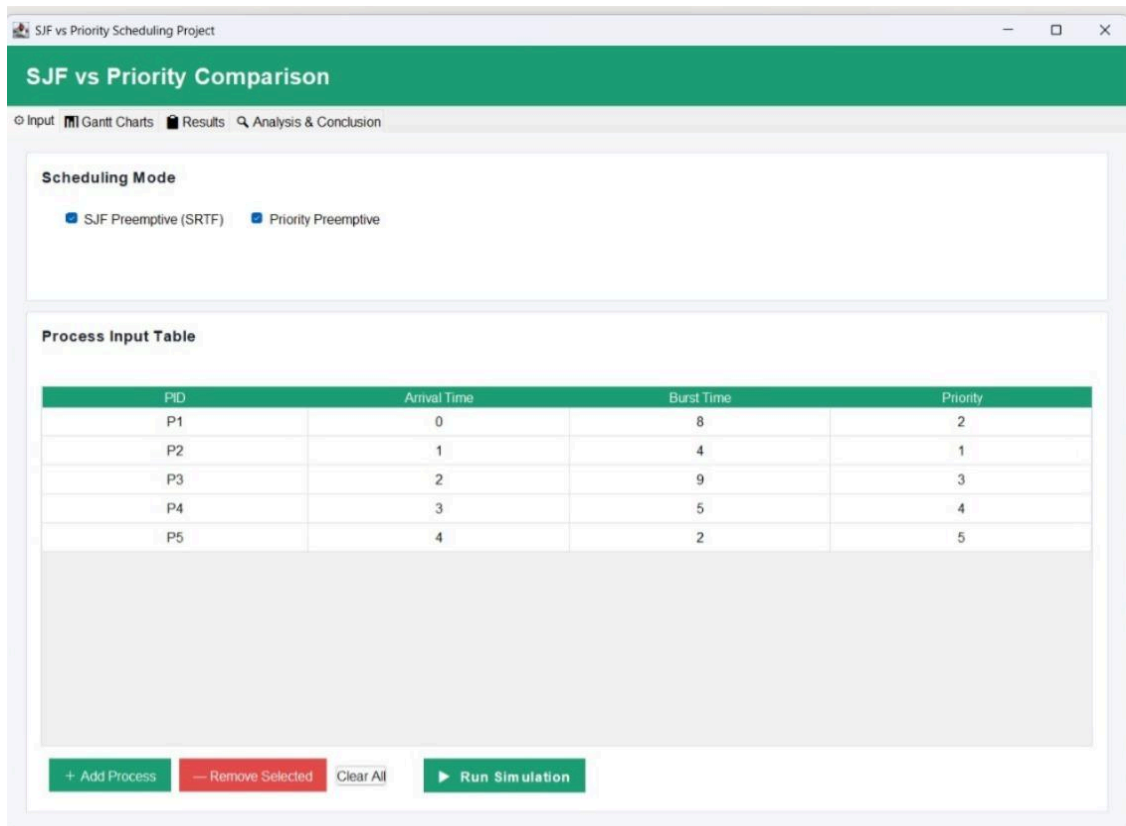
- SJF max single-process wait: **17 units** | Priority: **22 units**
- ✓ SJF: no starvation risk detected for this workload.
- ⚠ Priority: low-priority processes waited > 20 units — **starvation risk** without aging.

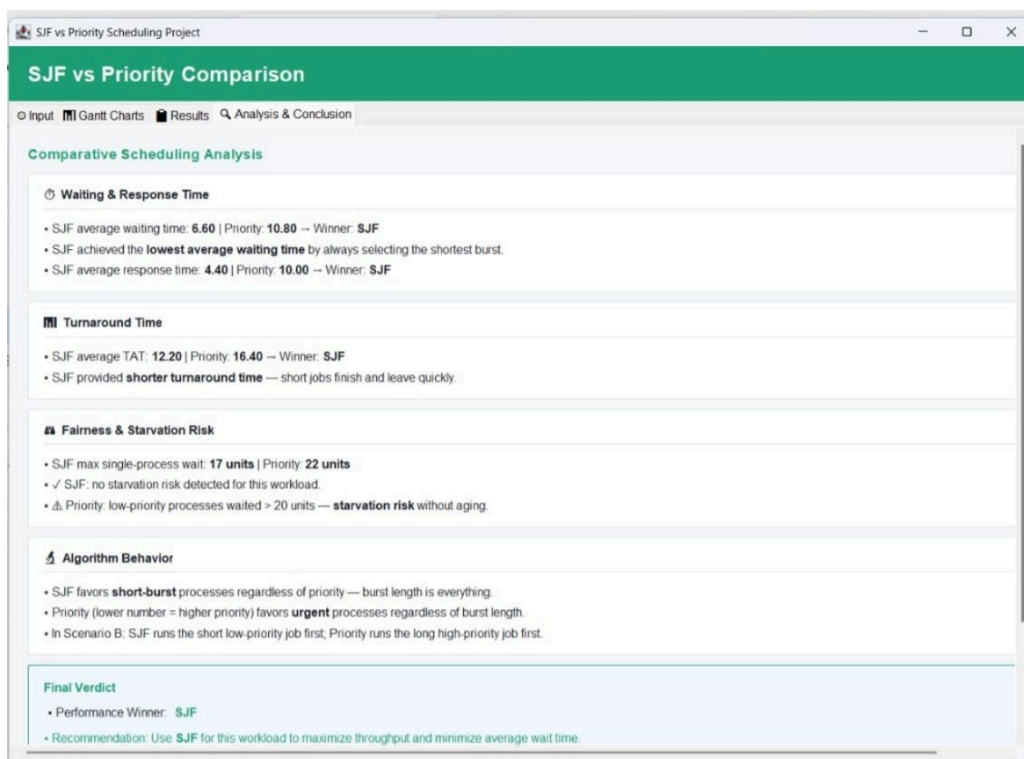
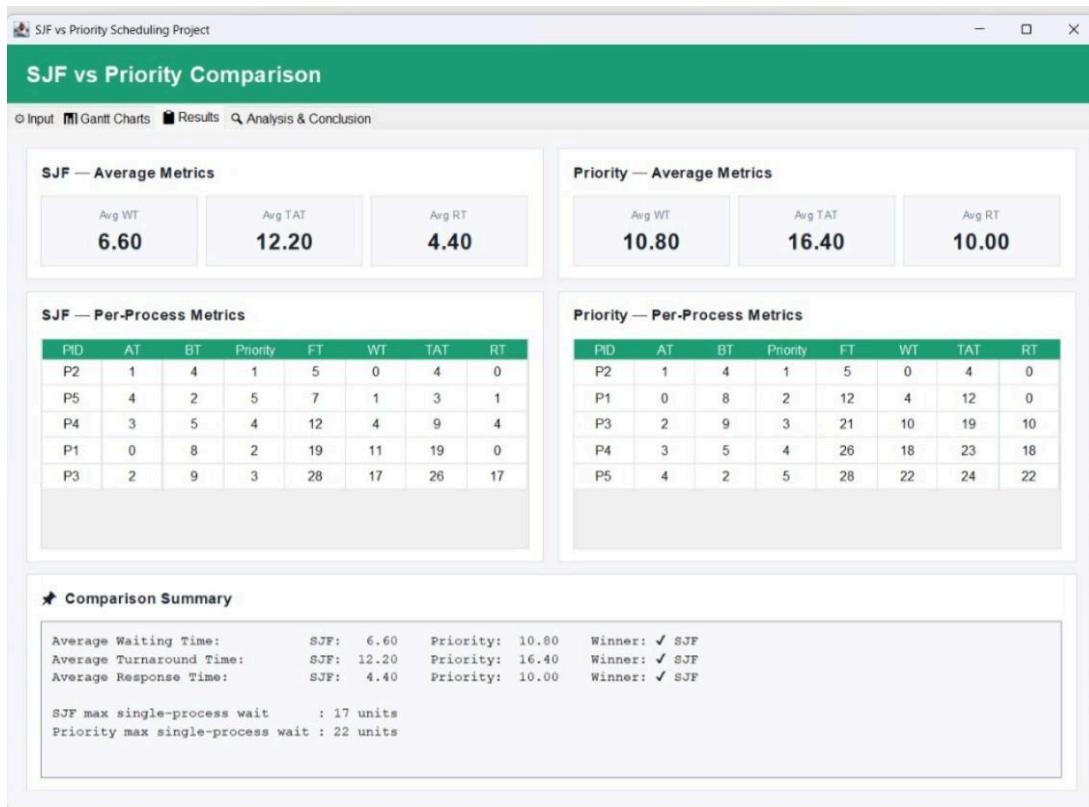
Algorithm Behavior

- SJF favors **short-burst** processes regardless of priority — burst length is everything.
- Priority (lower number = higher priority) favors **urgent** processes regardless of burst length.
- In Scenario B: SJF runs the short low-priority job first; Priority runs the long high-priority job first.

Final Verdict

- Performance Winner: **SJF**
- Recommendation: Use **SJF** for this workload to maximize throughput and minimize average wait time.





Scenario B: Conflict (Short-Burst vs. High-Priority)

- **Goal:** To observe the system's decision when a very short job has a low priority compared to a long job with high priority.
- **Analysis:** In Priority mode, the high-priority process took control first despite its length. In SJF, the short job was executed first to improve efficiency.
- **Conclusion:** This scenario highlights the sacrifice of system speed for the sake of task importance in Priority Scheduling.

SJF vs Priority Scheduling Project

SJF vs Priority Comparison

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Scheduling Mode

☐ SJF Preemptive (SRTF)

☐ Priority Preemptive

Process Input Table

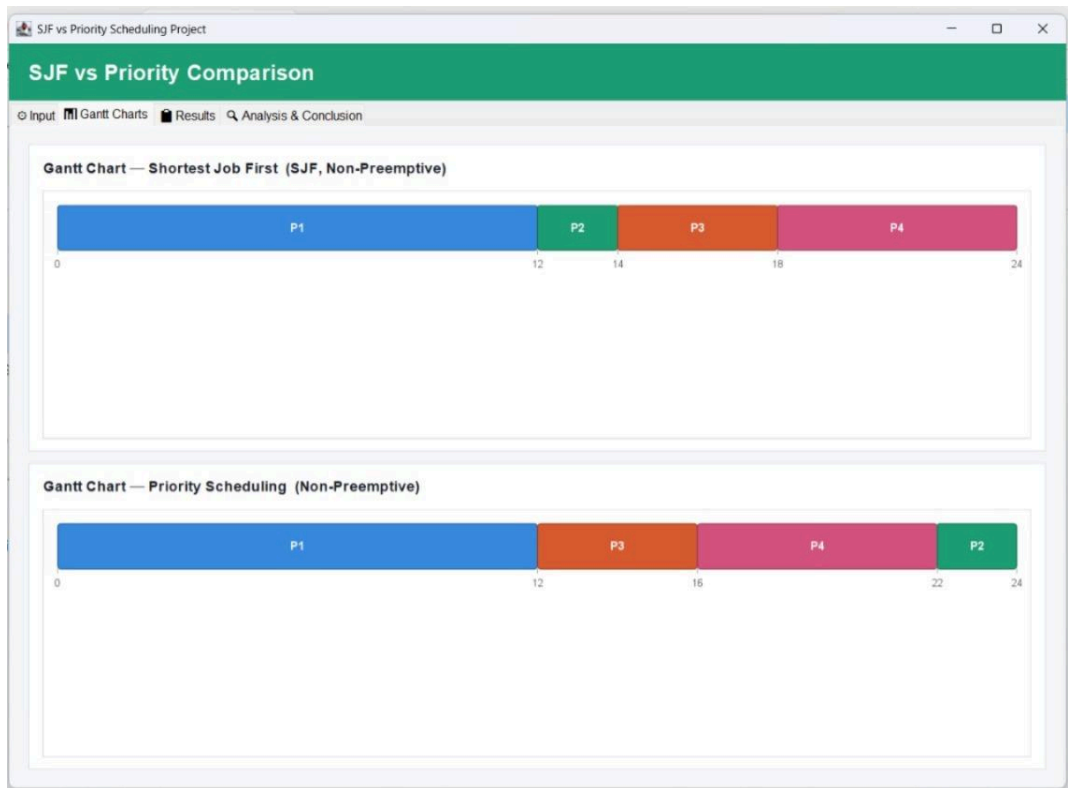
PID	Arrival Time	Burst Time	Priority
P1	0	12	1
P2	2	2	5
P3	3	4	2
P4	5	6	3

+ Add Process

— Remove Selected

Clear All

▶ Run Simulation



SJF vs Priority Scheduling Project

SJF vs Priority Comparison

Input Gantt Charts Results Analysis & Conclusion

SJF — Average Metrics

Avg WT	Avg TAT	Avg RT
8.50	14.50	8.50

Priority — Average Metrics

Avg WT	Avg TAT	Avg RT
10.00	16.00	10.00

SJF — Per-Process Metrics

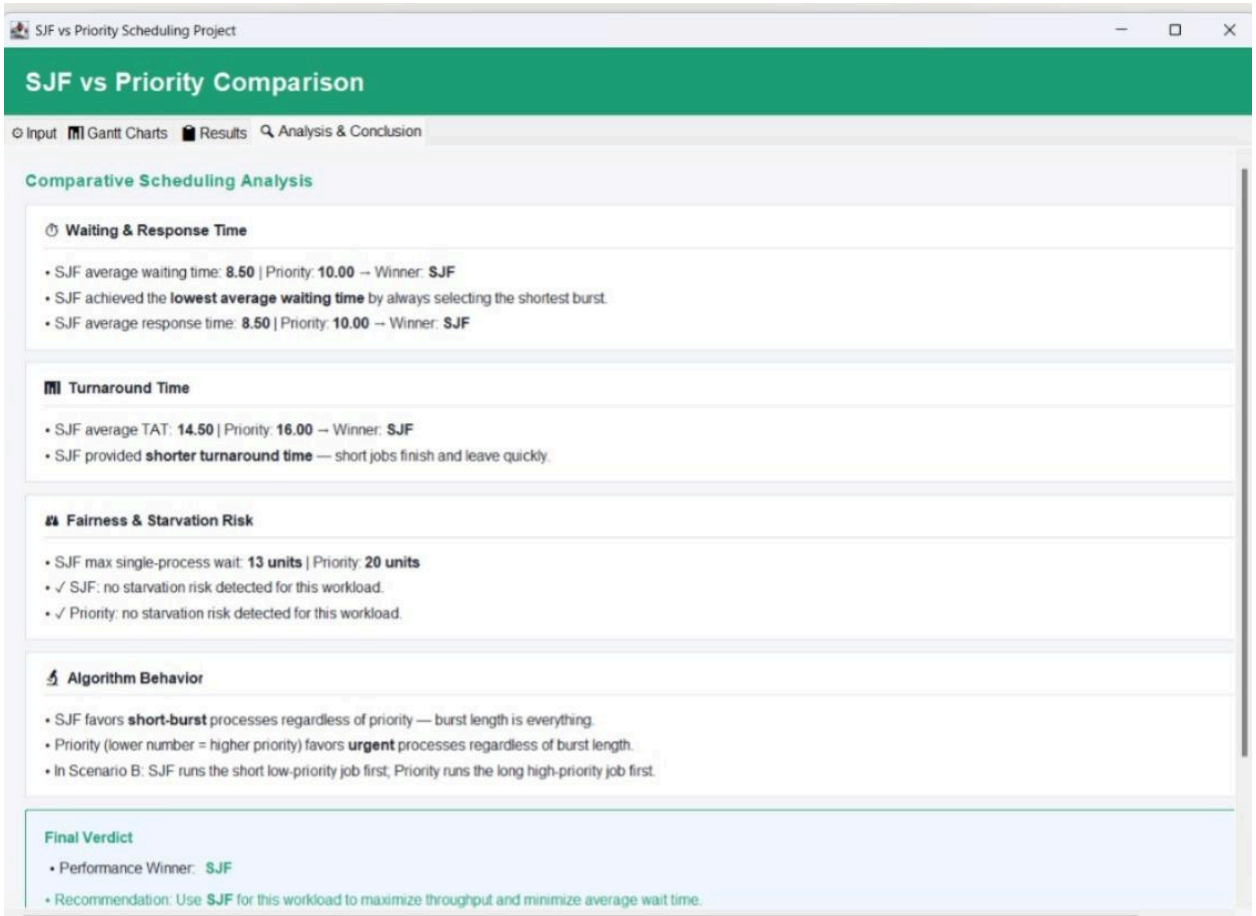
PID	AT	BT	Priority	FT	WT	TAT	RT
P1	0	12	1	12	0	12	0
P2	2	2	5	14	10	12	10
P3	3	4	2	18	11	15	11
P4	5	6	3	24	13	19	13

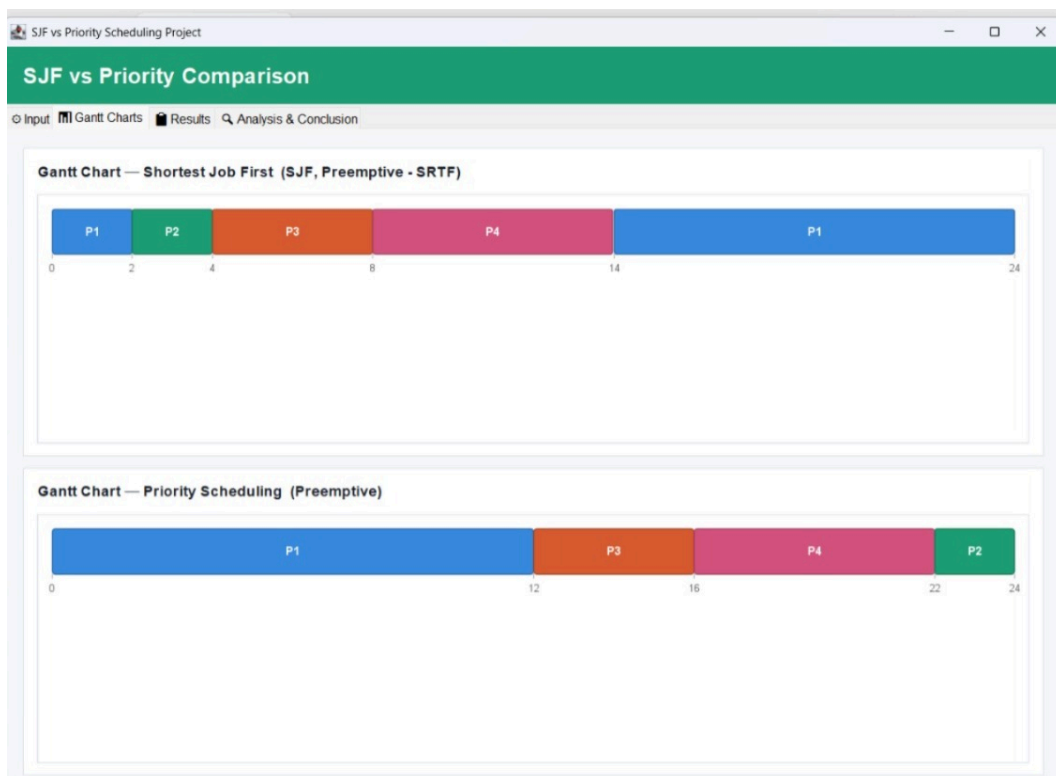
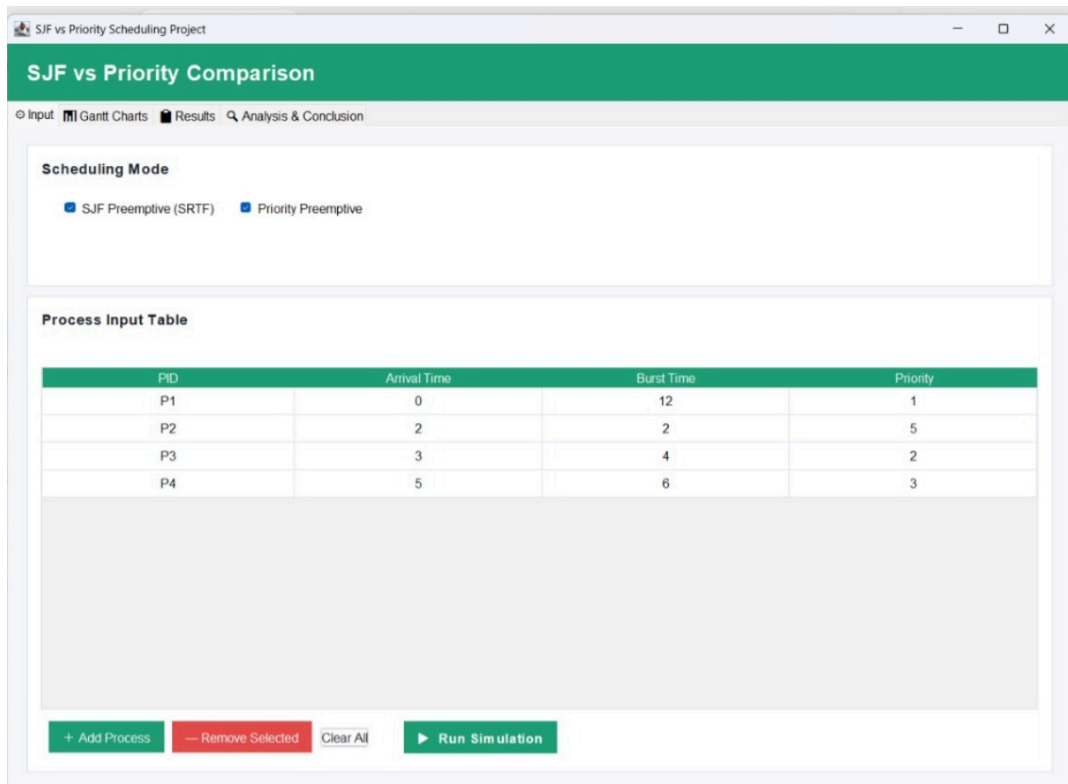
Priority — Per-Process Metrics

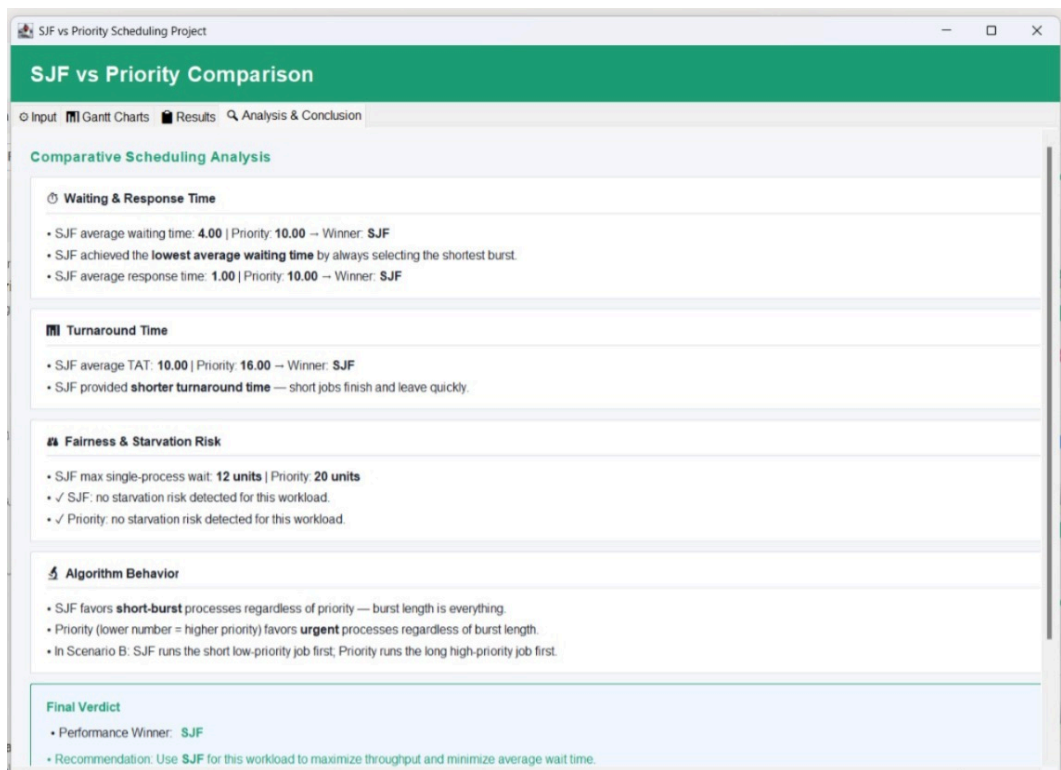
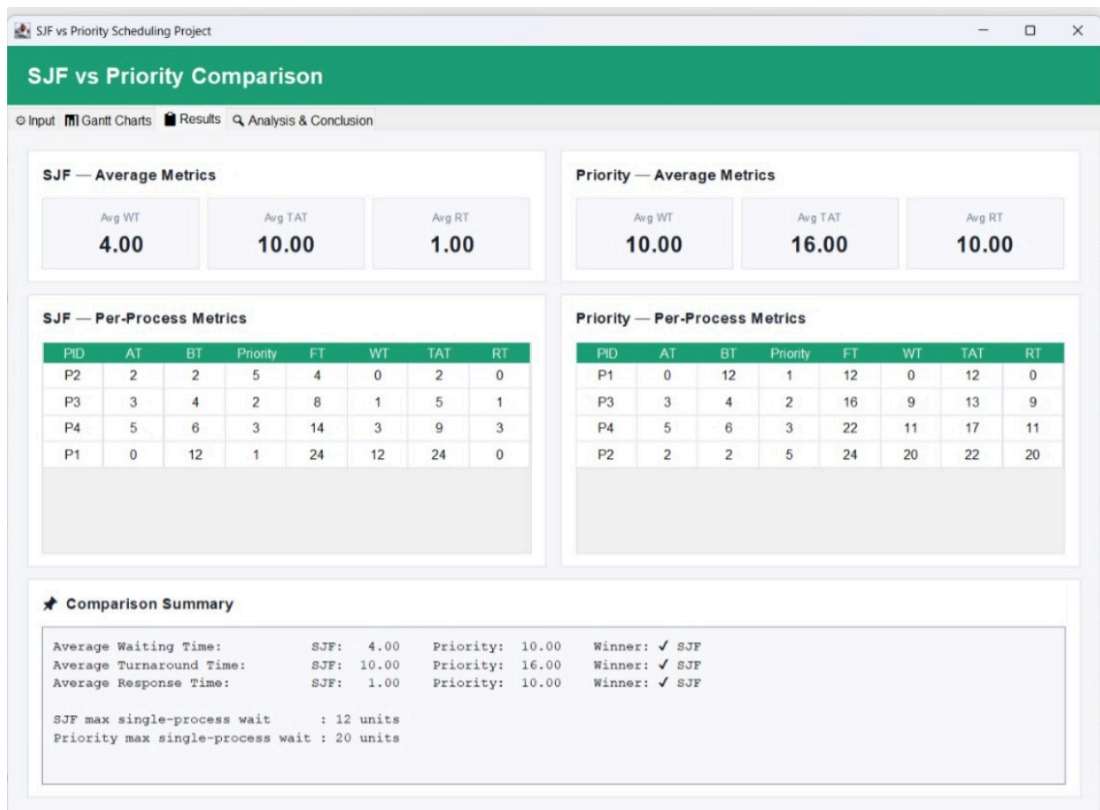
PID	AT	BT	Priority	FT	WT	TAT	RT
P1	0	12	1	12	0	12	0
P3	3	4	2	16	9	13	9
P4	5	6	3	22	11	17	11
P2	2	2	5	24	20	22	20

★ Comparison Summary

Average Waiting Time:	SJF: 8.50	Priority: 10.00	Winner: ✓ SJF
Average Turnaround Time:	SJF: 14.50	Priority: 16.00	Winner: ✓ SJF
Average Response Time:	SJF: 8.50	Priority: 10.00	Winner: ✓ SJF
SJF max single-process wait : 13 units			
Priority max single-process wait : 20 units			



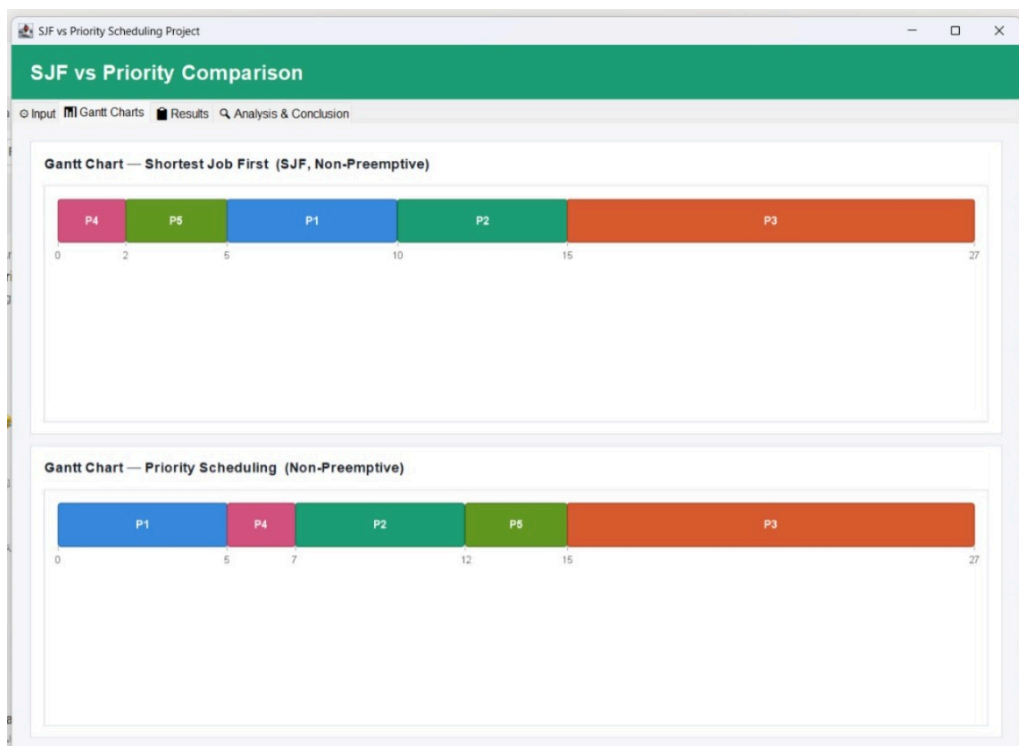
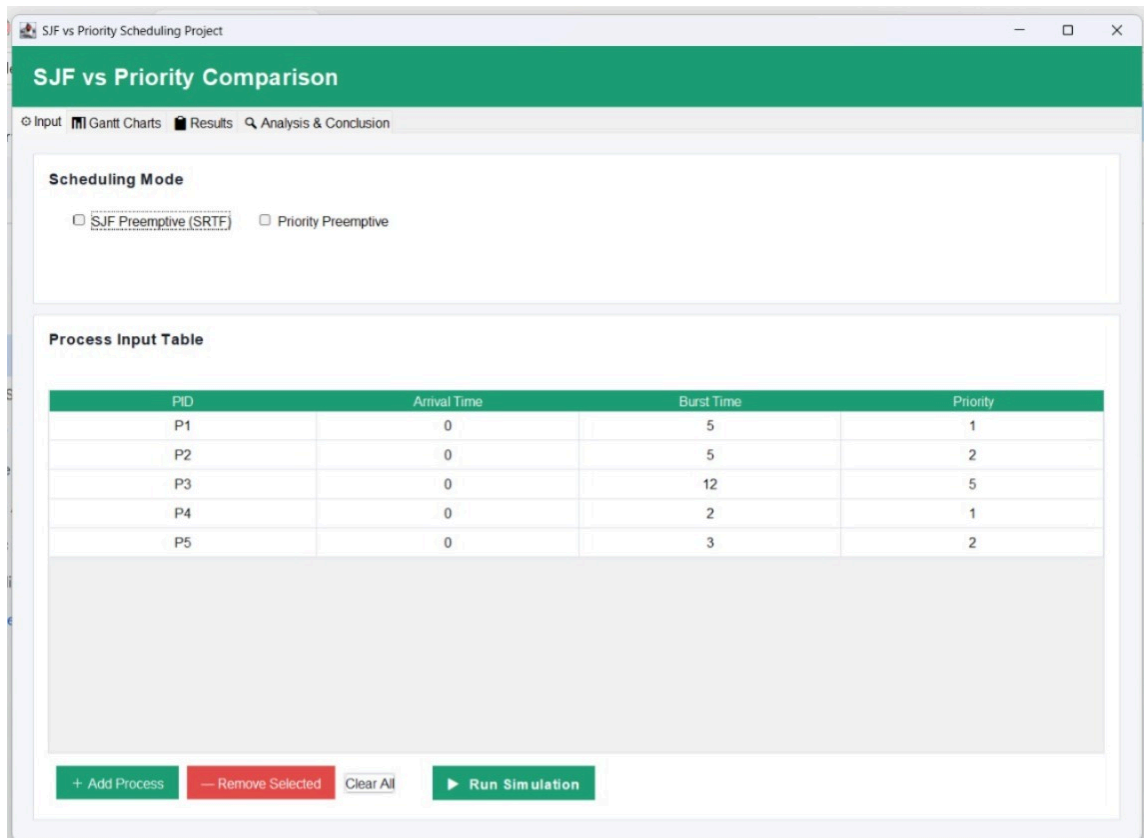


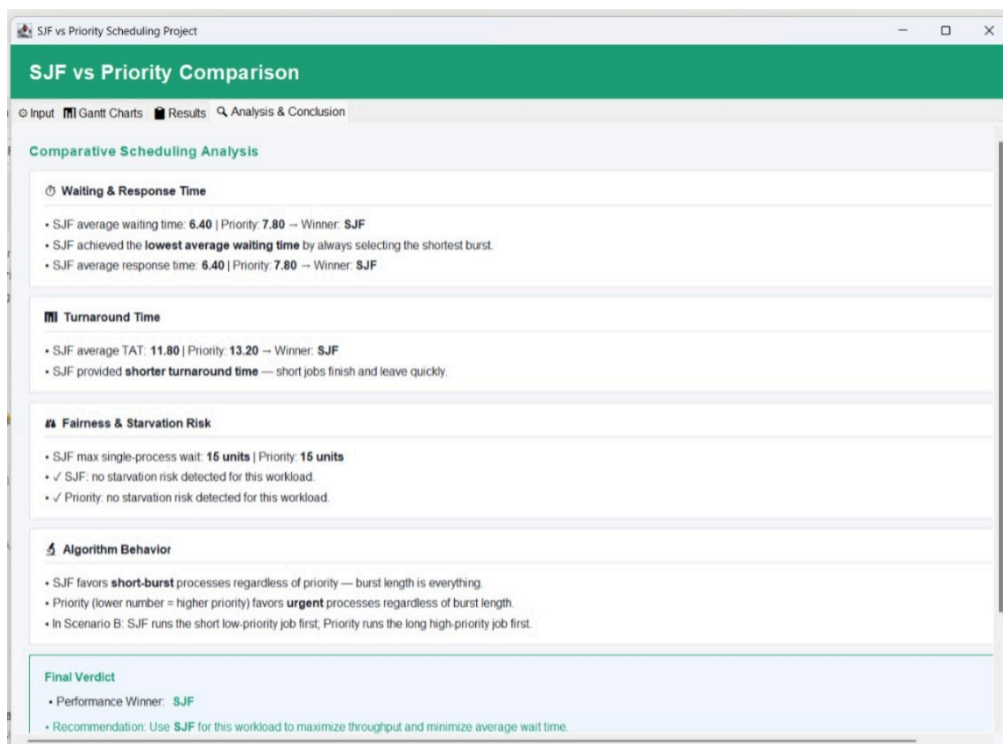
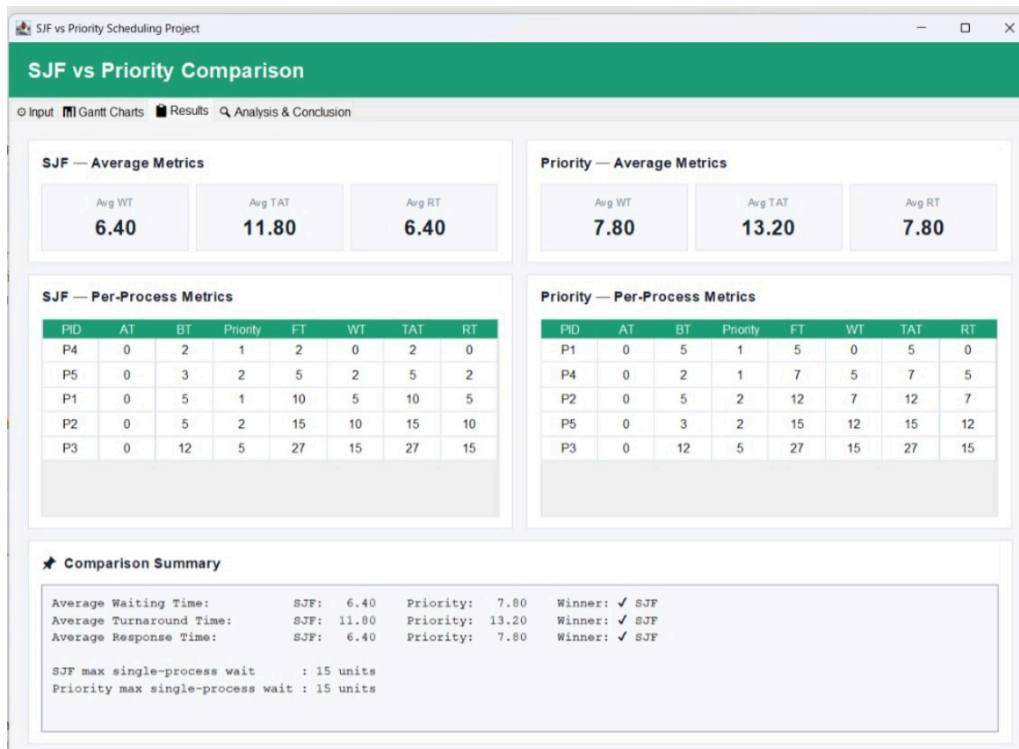


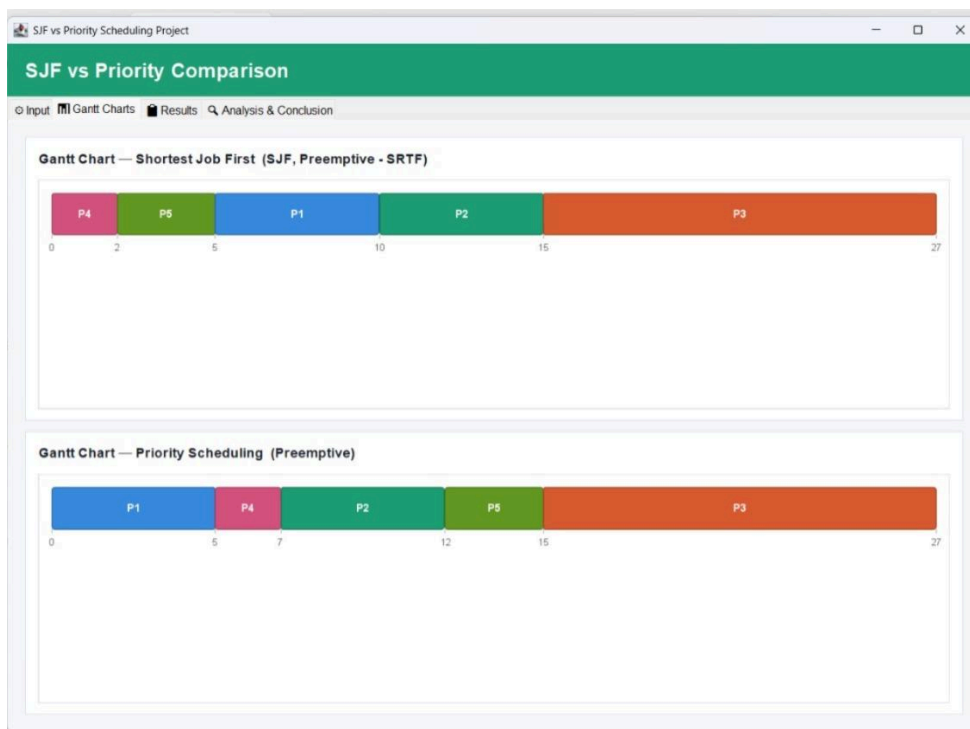
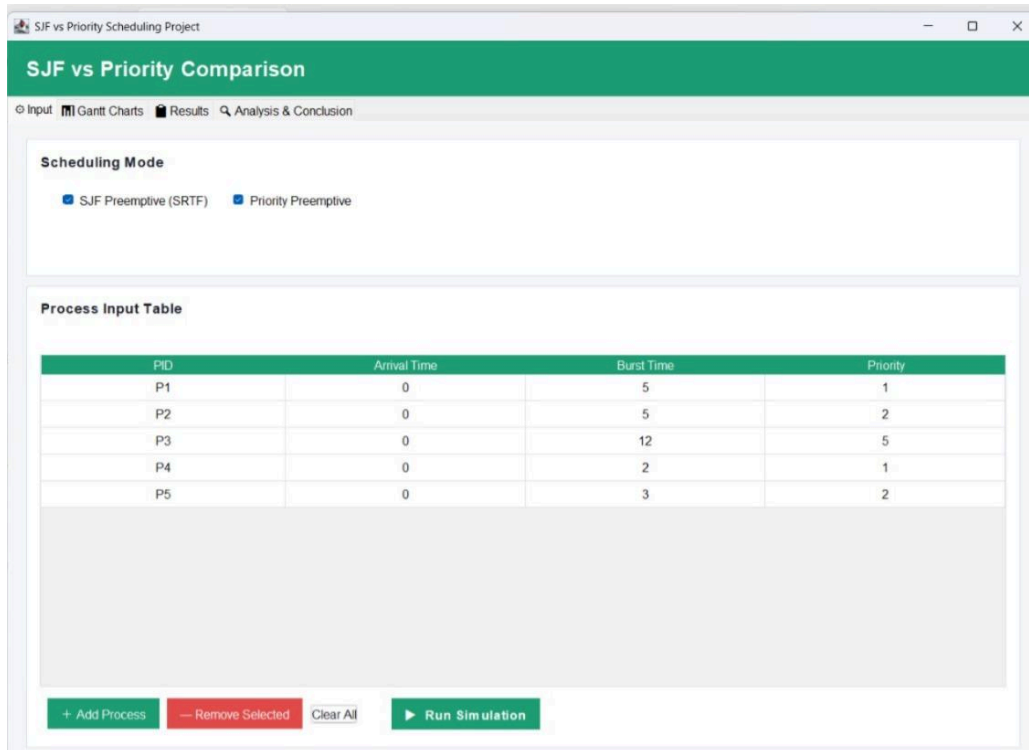
"In Scenario B, Priority Scheduling caused process **P2** to wait for a long time despite its small burst time, because it had a low priority. Conversely, SJF prioritized smaller tasks to keep the average waiting time low."

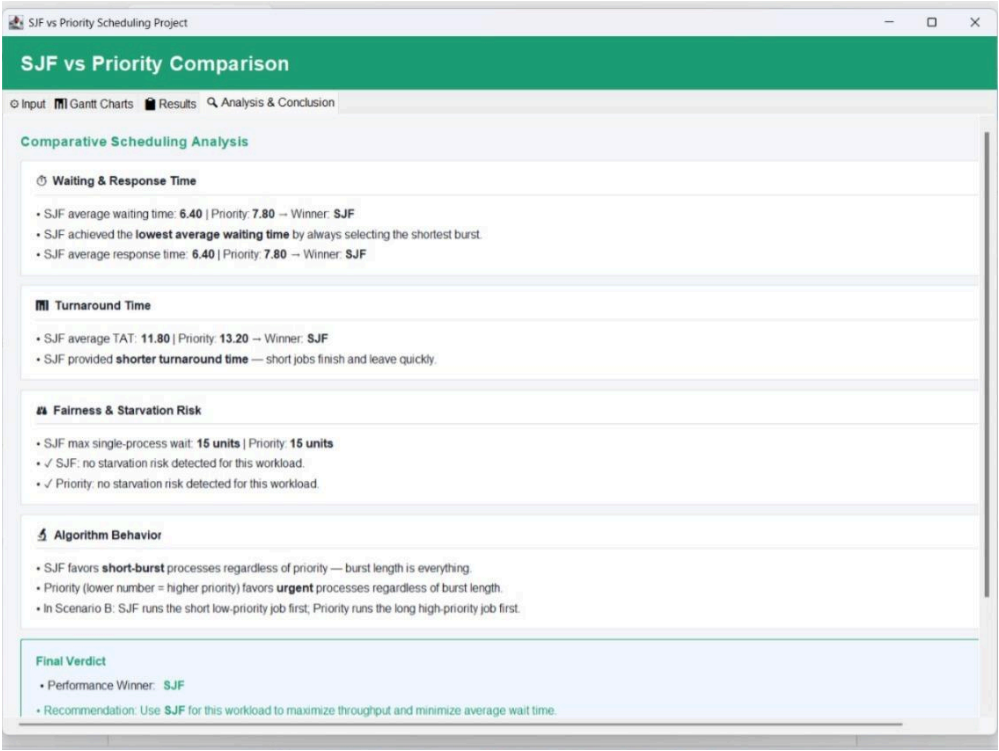
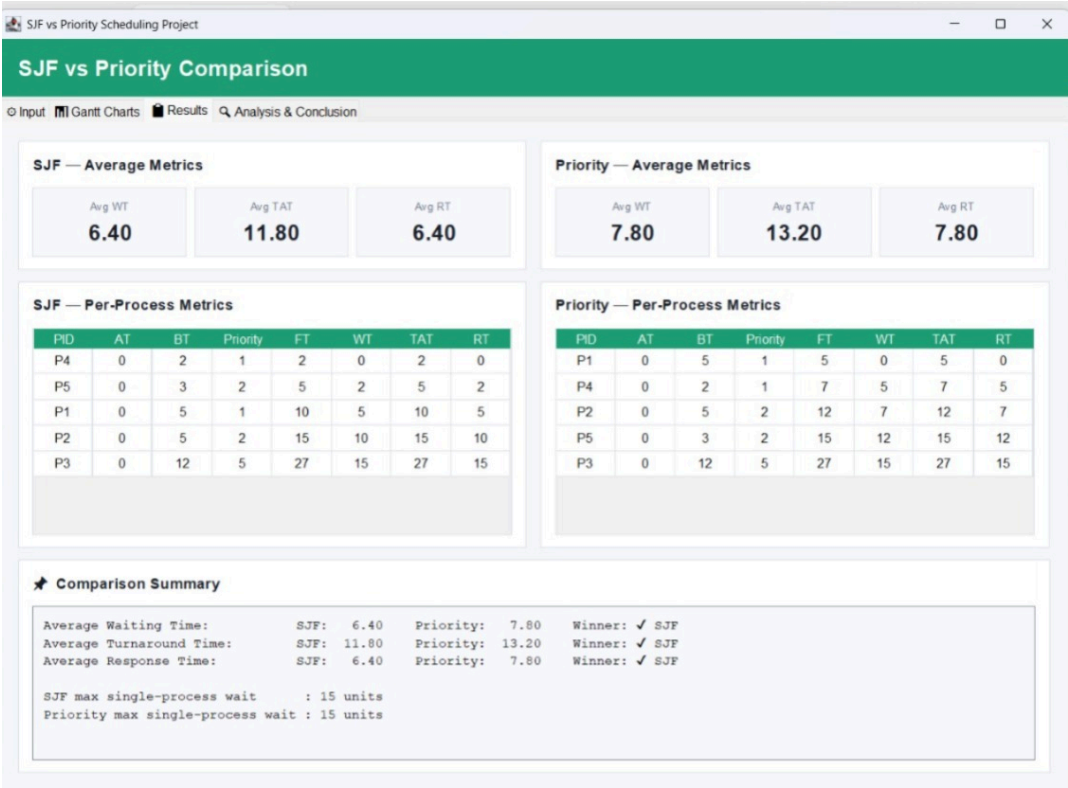
Scenario C: Fairness & Starvation Case

- **Goal:** To detect "Starvation" where a process is delayed indefinitely.
- **Analysis:** In Priority Scheduling, the process with the lowest priority (highest number) suffered a massive delay (Starvation). SJF was fairer as it eventually processed all jobs based on their size.
- **Conclusion:** Priority scheduling requires "Aging" techniques to be fair, whereas SJF is naturally more balanced for smaller tasks.









"Scenario C demonstrates the **Starvation** issue in Priority Scheduling. Process **P3** suffers from an unfair delay because it has the lowest priority, even though it arrived at the same time as others. SJF handled the workload based on task size, showing better overall fairness for the system."

Scenario D: Validation & Error Handling

- **Goal:** To ensure the robustness of the software against invalid user inputs.
- **Analysis:** When entering negative values or non-numeric characters (e.g., "abc"), the system successfully blocked the execution and displayed a clear error message.
- **Conclusion:** The application is robust and prevents system crashes by validating all inputs before processing.

The screenshot shows a web application titled "SJF vs Priority Scheduling Project". The main heading is "SJF vs Priority Comparison". Below this, there are tabs for "Input", "Gantt Charts", "Results", and "Analysis & Conclusion". The "Input" tab is active, showing a "Scheduling Mode" section with two radio buttons: "SJF Preemptive (SRTF)" and "Priority Preemptive". Below this is a "Process Input Table" with a warning message: "Process P3: Arrival Time must be > 0". The table has four columns: PID, Arrival Time, Burst Time, and Priority. It contains four rows of data for processes P1, P2, P3, and P4. At the bottom of the table, there are four buttons: "+ Add Process", "- Remove Selected", "Clear All", and "Run Simulation".

PID	Arrival Time	Burst Time	Priority
P1	0	2	1
P2	0	3	2
P3	-2	12	5
P4	0	2	1

SJF vs Priority Scheduling Project

SJF vs Priority Comparison

Input

Gantt Charts

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Scheduling Mode

☐ SJF Preemptive (SRTF)

☐ Priority Preemptive

Process Input Table

⚠ Process P2: Burst Time must be an integer (got "abc")

PID	Arrival Time	Burst Time	Priority
P1	0	2	1
P2	0	abc	2
P3	-2	12	5
P4	0	2	1

+ Add Process

— Remove Selected

Clear All

▶ Run Simulation

"This scenario tests the robustness of the application. The system successfully detects invalid inputs (such as non-numeric values or negative times) and prevents the simulation from running, providing clear feedback to the user."